Triangle Inequality Theorem; Converse of the Pythagorean Theorem

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lessons 1–2 — Triangle Inequality Theorem; Converse of the Pythagorean Theorem	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.3

2. Standards

MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.

Must. Decide whether three lengths make a triangle (the two shorter sum to more than the longest) and a right triangle (the converse, on the longest side).

Must not. Acute or obtuse by comparing squares (Geometry); sides as expressions; finding a missing side (6.03).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I: three lengths as sticks and as one inequality
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 3, 5 and 7 split; the sixty-second re-vote
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	board 7: a reason checked against the squares, not the sum

4. Learning Target and Essential Question

Learning target. I can decide whether three lengths can make a triangle, and whether they make a right triangle.

Essential question. Why can't any three sticks make a triangle, and how do three numbers tell you a corner is square?

Building on. Grade 7 (MA.7.NSO.1.1) — squares of whole numbers; 4.05 — square roots of perfect squares; Grade 5 (MA.5.GR.1.1) — a right triangle has one right angle.

Working toward. 6.03 — the Pythagorean Theorem finds a missing side; 6.05–06 — distance on the coordinate plane; Algebra 1 and Geometry — the Triangle Inequality in proofs.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	10	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Two sides of a triangle are 5 and 9. Why can the third side not be 4, and why can it not be 14?	DOK 3
Lengths 6, 8 and 10 make a right triangle. Without computing, do 12, 16 and 20? Why?	DOK 2
A student tests $13^2 + 12^2 = 5^2$ for sides 13, 12 and 5 and decides it is not a right triangle. What should she have tested?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Can sides of 2 ft, 7 ft and 10 ft make a triangle?	***No — 2 + 7 = 9 is less than 10.***	Yes — added 7 + 10 or 2 + 10 [MA.8.GR.1.3: test the two shorter]; 'yes — the ends just do not meet' [Florida B1G-M, GR.1.3: believing the sides need not meet at the corners]
2	Can sides of 6 m, 6 m and 11 m make a triangle?	***Yes — 6 + 6 = 12 is more than 11.***	No — 'two sides cannot be equal' [Notes I: only the sum matters]; 'it is not a right triangle, so the rule does not apply' [Florida B1G-M, GR.1.3: the Triangle Inequality holds for every triangle]
3	Which three lengths can make a triangle?	***B — 5, 11, 8***	(A) 7 + 7 = 14 counted — equal lies flat [Notes I]; (C) added the first two listed, 3 + 10 [MA.8.GR.1.3: test the two shorter]; (D) added the longest with another, 12 + 4 [MA.8.GR.1.3: test the two shorter]
4	Two sides of a triangle are 7 cm and 12 cm. Write an inequality for the length x, in centimeters, of the third side.	***5 < x < 19 — more than 5 cm and less than 19 cm.***	5 ≤ x ≤ 19 — the flat cases included [Notes II: never equal]; x < 19 — no bottom end [Notes II: long enough to reach]
5	Two sides of a triangle are 6 cm and 10 cm. Which inequality shows every possible length x, in centimeters, of the third side?	***C — 4 < x < 16***	(A) between the two given sides — the third side can be shorter than both [Notes II: more than the difference]; (B) the flat cases included: at 4 or 16 the sides lie flat [Notes II: never equal]; (D) no bottom end — a third side of 4 or less cannot reach [Notes II: more than the difference]
6	Do sides of 8 inches, 15 inches and 17 inches make a right triangle?	***Yes — $8^2 + 15^2 = 289 = 17^2$.***	No — compared 8 + 15 = 23 with 17 [Notes III: squares, not sides]

#	What the board asks	Answer	If the board is wrong
7	A triangle has sides of 25 cm, 7 cm and 24 cm. Is it a right triangle? Which answer has the right reason?	***A — yes, because $7^2 + 24^2 = 25^2$***	(B) the right verdict for the wrong reason — the Triangle Inequality shows only a triangle [MA.8.GR.1.3: two tests]; (C) 25 taken as a leg [Florida B1G-M, GR.1.1: misidentifying the hypotenuse]; (D) added the sides, not their squares [Notes III]
8	A triangular sail has sides of 20 ft, 21 ft and 29 ft, and it weighs 12 pounds. Does the sail have a right angle?	***Yes — $20^2 + 21^2 = 841 = 29^2$.***	'Cannot tell without the angles' [Notes III: the sides are enough]
9	Ana says sides of 6, 7 and 9 make a right triangle because 6 + 7 > 9. Is she right? Explain.	***No — 6 + 7 > 9 shows only a triangle; 85 is not 81, so it is not right.***	'Yes, 13 > 9' — the inequality taken as the converse [MA.8.GR.1.3: two tests]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Two lines stay on the board all period: "the two SHORTER sides, added, beat the longest" and "the two shorter sides' squares, added, EQUAL the longest side's square". Every board starts by circling the longest side. Popsicle-stick sets for Notes I if a picture is not enough.

ELL. **Inequality** means "not equal" — here, MORE than. **Hypotenuse**: the long side across from the square corner. Say "longest side" with every board, and point to it.

Enrichment. Board 4 finishes early for some: how many whole-number lengths can the third side have when the other two are 7 and 12? (Thirteen: 6 through 18.) Then: which of them make a right triangle? (None is a whole number: √95 and √193.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: Ana's test (6 + 7 > 9) proves only that the three sides make a triangle; the converse is a test of squares, and 85 is not 81. A board that says 'she is wrong' without the squares has the verdict and not the reason.

Vocabulary. **triangle inequality**; **legs**; **hypotenuse**; **converse**

Materials. Whiteboards and markers. Optional: strips cut to 3, 4, 5, 6 and 9 units for Notes I. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Triangle inequality — 2N5; Converse of the Pythagorean theorem: is it a right triangle? — EQZ.